MIL-M-00050079E(MU)
16 December 1968
USED IN LIEU OF
MIL-M-50079D
30 September 1964

MILITARY SPECIFICATION

MASK, PROTECTIVE, FIELD ABC-M17

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-50079D. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers one type of protective mask with filter elements in the facepiece.

1.2 Classification. Masks shall be of the following sizes, as specified (see 6.2):

Small
Medium
Large

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-530	-	Plywood, Flat-Panel.
QQ-S-781	-	Steel Strapping, Flat.
TT-W-572	-	Wood-Preservative, Water Repellent
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Fiberboard.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).

FSC 4240

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MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-131	-	Barrier Material; Water Vaporproof, Flexible.
MIL-B-13239	-	Barrier Material, Waterproofed, Flexible, All Temperature.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MS24535	-	Fastener, Container Closure.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL5-1-293	-	Mask, Protective, Field, ABC-M17.
C5-1-545	-	Palletization of Mask, Protective, Field, ABC-M17.
A18-60-757	-	Interlocking, Slotted Partitions.
DL136-14-62	-	Q43 Inhalation and Exhalation Resistance Indicator.
DL136-42-102	-	Q127 DOP Filter Testing Penetrometer.
DL136-42-750	-	M14 Gas Mask Leakage Tester.
DL136-45-202	-	M4A1 (Mod.) Outlet Valve Leakage Indicator.
D150-2-1	-	Marking Diagram for Shipping Containers.

PUBLICATIONS

EDGEWOOD ARSENAL PURCHASE DESCRIPTION

197-54-1021	-	Filter Element, Protective Mask, M13A1.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-1-293 or its subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawing.

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3.1.2 Components. All components shall conform to the specifications and drawings listed on Drawing DL5-1-293 and its subsidiary drawings.

3.2 Assembly. The mask shall be assembled in accordance with Drawing E5-1-293.

3.2.1 Filter elements. Only matched pairs of filter elements shall be assembled into finished masks (see PD 197-54-1021).

3.3 Leakage.

3.3.1 Mask assembly leakage. Leakage of the mask assembly (excluding leakage at the periphery, outlet-valve seat, and through filter elements) shall not exceed 0.003 percent dioctylphthalate (DOP) smoke penetration when tested as specified in 4.2.4.1.

3.3.2 Outlet-valve leakage. Leakage of the outlet valve shall not exceed 15 milliliters of air per minute at a pressure differential of 25 millimeters of water when tested as specified in 4.2.4.2.

3.3.3 Package leakage and heat seal. Package leakage and heat seal tests shall be conducted on packaged mask assemblies as specified in 4.2.4.7.

3.4 Resistance.

3.4.1 Inhalation resistance. Inhalation resistance of the mask assembly shall not exceed 55 millimeters of water at a continuous airflow of 85 liters per minute when tested as specified in 4.2.4.3.

3.4.2 Exhalation resistance. The exhalation resistance of the mask assembly shall not exceed 23 millimeters of water at a continuous airflow of 85 liters per minute when tested as specified in 4.2.4.4.

3.5 Filter elements.

3.5.1 Moisture content of filter elements in mask. The average moisture content of the filter elements in the packaged mask shall not exceed 2.0 percent and no individual element shall exceed 2.5 percent moisture when tested as specified in 4.2.4.5.

3.5.2 DOP smoke penetration. After moisture content testing is completed, the same filter elements shall be tested for DOP smoke penetration. The DOP smoke penetration of each filter element shall not exceed 0.015 percent when tested as specified in 4.2.4.6.

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3.6 Workmanship. The mask assembly shall be free from foreign matter, and damage such as permanent set of rubber parts, cracks, dents, tears, and malfunctioning of hardware.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the masks of one size produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions. However, no more than one lot of filter elements, faceblanks, voicemitter outlet-valve assemblies, or nosecups shall be represented in any one lot of finished masks.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For destructive tests. Sampling for 4.2.4.5 and 4.2.4.6 shall be conducted in accordance with MIL-STD-105, using level S-3. Sampling for 4.2.4.7 shall be conducted in accordance with MIL-STD-105, using level S-4.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive tests. Sample masks from each lot shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

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4.2.3.2 For destructive tests. Samples taken in accordance with 4.2.2.2 shall be tested as specified in 4.2.4.5, 4.2.4.6 and 4.2.4.7. The acceptance number is 0.

4.2.3.3 For critical defects. Each mask in each lot shall be inspected for critical characteristics listed in the classification of defects.

4.2.3.4 Classification of defects.

(a) Mask assembly (Dwg E5-1-293).

Categories	Defects	Acceptance standards
Critical:		
1	Mask assembly leakage	4.2.4.1
2	Outlet valve leakage	4.2.4.2
3	Lens surface split or cracked	
Major:	AQL 0.25 percent defective	
101	Inhalation resistance	4.2.4.3
102	Exhalation resistance	4.2.4.4
	AQL 1.0 percent defective	
103	Component missing, incorrect, or incorrectly assembled	
104	Workmanship (3.6)	
105	Inlet valve assembly inoperable	
106	Marking	
107	Unmatched pair of filter elements	
108	Headharness buckle not free to rotate in loop of clip	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing or incorrect	
202	Marking missing or illegible	

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4.2.4 Tests.

4.2.4.1 Mask assembly leakage. Leakage of the mask assembly, with filter elements plugged or capped (less inlet valve assemblies) and with outlet valve opening plugged, shall be determined with the M14 Gas Mask Leakage Tester (DL136-42-750). Each eyepiece assemoly and voicemitter assembly shall be flexed by hand during the testing operation.

4.2.4.2 Outlet valve leakage. The leakage of the outlet valve shall be determined with the M4A1 (Mod.) Outlet Valve Leakage Indicator (DL136-45-202).

4.2.4.3 Innalation resistance. Innalation resistance of the mask assembly shall be determined with the Q43 Inhalation and Exhalation Resistance Indicator (DL136-14-62).

4.2.4.4 Exhalation resistance. Exhalation resistance of the mask assembly shall be determined with the Q43 Innalation and Exhalation Resistance Indicator (DL136-14-62).

4.2.4.5 Moisture content of filter elements. The filter elements from each sample mask shall be weighed to the nearest 0.1 gram (wet weight) immediately after unpacking the mask. The elements shall be placed in a forced-air drying oven at 160° ± 5°F. and a maximum of 5 percent relative humidity until constant dry weight, within 0.1 gram, is reached. The final weighing (dry weight) shall be recorded to the nearest 0.1 gram and the percent moisture content determined by the following formula:

$$\text{Percent moisture content} = \frac{100\ (\text{wet weight} - \text{dry weight})}{(\text{wet weight}}$$

4.2.4.6 DOP smoke penetration. DOP smoke penetration of the filter elements shall be determined with the Q127 DOP Filter Testing Penetrometer (DL136-42-102) at a flow-rate of 16 liters per minute.

4.2.4.7 Package leakage and heat seal test. Leakage and heat seal characteristics of the sealed water-vaporproof bag containing the mask assembly shall be determined by subjecting the samples taken in accordance with 4.2.2.2 to the hot water technique and heat seal seam tests of MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A.

5.1.1 Type I mask packaging (see 6.1.2 and 6.2). No preservative compounds shall be applied to any portion of the mask or its components.

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One M1 waterproof bag shall be put in the same pocket of the canvas carrier provided for the M13 Individual Decontaminating and Reimpregnating Kit. Each mask, with a faceform positioned in the mask as shown on Drawing E5-1-293, shall be placed in a water-vaporproof heat-sealable bag fabricated from barrier-material conforming to class 1 of MIL-B-131, except that the heat-sealable surface of the material shall be a nonplasticized type, such as polyethylene. The package shall be in conformance with the requirements of method IA-8 of MIL-P-116, except that sampling shall be conducted in accordance with 4.2.2.2 of this specification. Water-vaporproof bags cited above shall have the same inspection acceptance criteria as the packaged mask (see 4.2.3.2). Cushioning, shall be nonhydroscopic, nondusting, and compatible with the mask. Each packaged mask shall be placed in a canvas carrier as shown on Drawing E5-1-293. The canvas carrier containing the packaged mask shall then be placed in a W5c or W6c, style RSC, fiberboard box, conforming to PPP-B-636, having an approximate inside dimension of 7-1/2 by 5-1/2 by 10-3/4 inches. In addition to closure requirements for slotted style interior boxes the box shall be taped as specified, except that pressure-sensitive tape conforming to PPP-T-76 shall be used.

5.1.2 Type II mask packaging (see 6.1.2 and 6.2). The mask shall be packaged in the same manner as specified in 5.1.1, except that the fiberboard box shall be omitted. In lieu thereof, each canvas carrier containing a packaged mask shall be placed in a waterproof heat-sealable bag fabricated from barrier-material, conforming to type B2 (stretchable or nonstretchable), class 1, grade A or B; or type B-3, class 1, grade A or B of MIL-B-13239. The package shall be in conformance with the requirements of method IC-3 of MIL-P-116, except that the package leakage and heat seal test need not be performed.

5.2 Packing, level A.

5.2.1 Type I fiberboard pack (see 6.1.2). Ten boxes of masks, of one size, packaged as specified in 5.1.1, shall be packed in a V3c, style RSC, fiberboard box conforming to PPP-B-636. The box shall be closed and taped as specified in the appendix to PPP-B-636. Tape banding shall be used in lieu of metal banding to reinforce box.

5.2.1.1 Type II plywood pack (see 6.1.2). Ten boxes of masks, of one size, packaged as specified in 5.1.1, shall be packed in an overseas type, style A, cleated-plywood box conforming to PPP-B-601 and using plywood complying to group A of NN-P-530 in accordance with type I, grade 3 or 4 of CS-35, provided the plywood does not have continuous holes or openings extending the entire panel from face to back. The box shall be closed and strapped in accordance with the appendix thereto. Strapping conforming to type I, class B of QQ-S-781 shall be used. The boxes or the finished wood parts thereof shall be completely immersed for a minimum of three minutes in a water repellent wood preservative conforming to composition A of TT-W-572.

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5.2.1.2 Type III garrison pack (see 6.1.2). Ten masks, of one size, packaged as specified in 5.1.2, shall be packed in a snug fitted V3c, style FTC fiberboard box conforming to PPP-B-636, without box cover. The box shall be divided into ten cells of the same size extending the full depth of the box by means of interlocking slotted partitions (Drawing A18-60-757) made from the same material as the box; the extended ends of the partitions shall be stapled or stitched to the sides, ends, and bottom of the box with staples or wire in accordance with the requirements of PPP-B-636 for the securing of the manufacturer's joint. The interlocking partitioned fiberboard box with the packaged masks packed therein, shall be placed in snug fitting, overseas type, style A, cleated-plywood box conforming to PPP-B-601, using type II, class 2, plywood conforming to NN-P-530. The cover of the cleated-plywood box shall not be nailed in place, but shall be secured to the box with cleated-box fasteners conforming to MS24535, using two fasteners on each side and one fastener at each end of the box. The fasteners shall be assembled to the box in accordance with the instructions of the fastener manufacture. In addition the cleated-plywood box shall be strapped with flat galvanized metal strapping in conformance with the appendix to PPP-B-601. Strapping conforming to type I, class B of QQ-S-781 shall be used. The cleated-box fasteners on the sides of the box shall be placed over the strapping.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with D150-2-1 (Marking Diagram for Shipping Containers). All markings on the shipping container shall be clear, legible and permanent after the application of the wood preservative specified in 5.2.1.1.

5.3.1 Marking for type III pack. The following additional markings shall be stencilled in minimum 3/4-inch high letters on the underside of the box cover:

BOX IS FOR USE IN STORING MASKS
BY GARRISON OR FIELD PERSONNEL

5.4 Repair parts. When applicable components for this mask are procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by the Federal Stock Number.

5.5 Palletization. Masks packed in accordance with 5.2.1.1 shall be palletized in accordance with Drawing C5-1-545. When the quantity of boxed masks is insufficient to constitute a full pallet load, the boxed masks shall not be palletized.

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6. NOTES

6.1 Intended use.

6.1.1 Mask. The mask covered by this specification is intended for protection of the eyes and respiratory system against toxicological warfare agents. It is not intended for protection against carbon monoxide or ammonia.

6.1.2 Packaging and packing (see section 5).

6.1.2.1 Type I. Type I package and pack is for shipment to areas where it is anticipated that hazards will be less extreme than in areas to which items will be shipped in type II containers.

6.1.2.2 Type I package and type II pack. Type I package and type II pack is intended for shipments to areas where extreme hazards, including outdoor exposure for long periods of time, are anticipated for handling, shipment, and storage. Masks shall be palletized (see 5.5).

6.1.2.3 Type II packaging and type III pack. Type II packaging and type III packing has been designated for use (and garrison of field reuse, when required) as a dual purpose shipping containers and as a device for the protection and order by garrison (or field) storage of masks during periods when masks which will have been issued to troops are not carried by them (see TM 3-4240-202-25P and TM 3-4240-202-15, Mask, Protective, Field, ABC-M17).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Size of mask.

(c) Elements for lot numbering (see Directive 511).

(d) Type packaging and packing.

6.3 Components specifications.

Title	Number
Backing Material, C18R1	MIL-B-51037
Bag, Waterproof, CBR Mask, M1	MIL-B-51170
Cap-Inlet Valve Assembly, C20R1	MIL-C-50081
Carrier. Field Protective Mask, M15A1	MIL-C-51285
Cloth, Marquisette, Polyester	MIL-C-51232
Core Material, C18R1	MIL-C-51096
Cover, Voicemitter-Outlet Valve, C13	MIL-C-50072

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Title	Number
Charcoal, Activated Impregnated, ASC	MIL-C-13724
Discs, Inlet and Nosecup Valve	MIL-D-50082
Discs, Outlet Valve	MIL-D-10132
Edgeseal Material	MIL-E-51065
Eyering, Field Protective Mask, C4	MIL-E-50069
Faceblank, C17R1	197-54-1020
Faceform, Field Protective Mask, M8	MIL-F-50073
Filter Element, Protective Mask, M13A1	197-54-1021
Filter Material, Gas-Aerosol, C18R1	MIL-F-51095
Head harness, C8R1	MIL-H-10120
Lenses, Ophthalmic, Simple (For ABC-M17 Mask)	MIL-L-50064
Nosecup, C10	197-54-1022
Outsert, Eyelens, M1	MIL-O-50077
Ring, Crimping (For ABC-M17 Mask)	MIL-R-50068
Valve, Nosecup	MIL-V-50065
Voicemitter-Outlet Valve, Assembly, C29R1	MIL-V-50066
Voicemitter-Outlet Valve, C29R1; Components for	MIL-V-50067

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement (see Serial No. 21, Supplement B to Quadripartite Standardization List Sheet Report). When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army - MU

International interest (see 6.4)

Preparing activity:

Army - MU(EA)

Project No. 4240-A381

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 1.9-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

MIL-M-0050079E

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-0270

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